≡ Navigation Tabs

Display Files | Download Files | Data API

6ECH | pdb_00006ech

Pyruvate Kinase Isoform L-type with phosphorylated Ser12 (pS12) in complex with FBP

PDB DOI: https://doi.org/10.2210/pdb6ECH/pdb

Classification: TRANSFERASE
Organism(s): Rattus norvegicus
Expression System: Escherichia coli
Mutation(s): No

Deposited: 2018-08-07 **Released:** 2019-12-04
Deposition Author(s): Padyana, A., Tong, S.

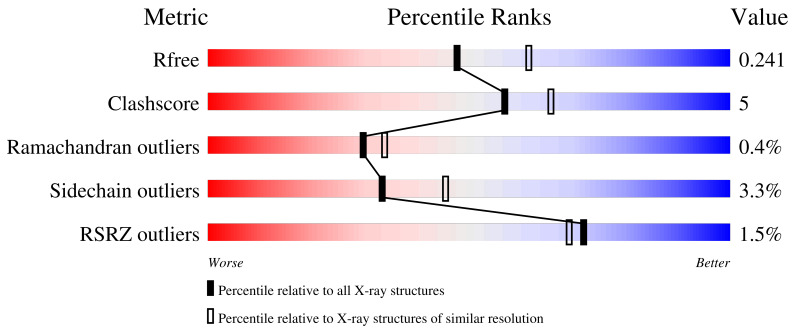
Experimental Data Snapshot

Method: X-RAY DIFFRACTION
Resolution: 2.19 Å
R-Value Free:
0.238 (Depositor), 0.241 (DCC)
R-Value Work:
0.181 (Depositor), 0.189 (DCC)
R-Value Observed:
0.184 (Depositor)

Starting Model: experimental
View more details

wwPDB Validation

3D Report | Full Report



Ligand Structure Quality Assessment



This is version 1.4 of the entry. See complete history.

Literature

Distinct Hepatic PKA and CDK Signaling Pathways Control Activity-Independent Pyruvate Kinase Phosphorylation and Hepatic Glucose Production.

Gassaway, B.M., Cardone, R.L., Padyana, A.K., Petersen, M.C., Judd, E.T., Hayes, S., Tong, S., Barber, K.W., Apostolidi, M., Abulizi, A., Sheetz, J.B., Aerni, H.R., Gross, S., Kung, C., Samuel, V.T., Shulman,

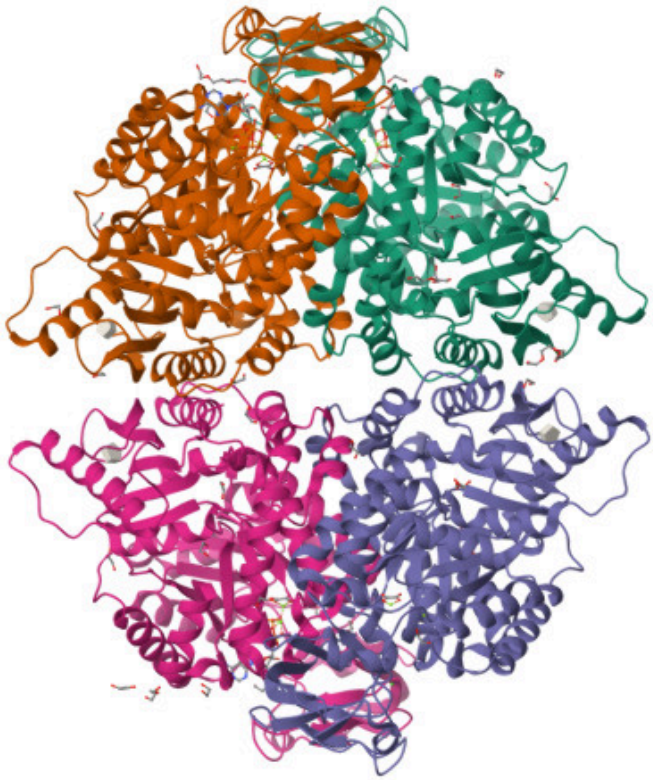
G.I., Kibbey, R.G., Rinehart, J.
(2019) Cell Rep **29**: 3394-3404.e9

PubMed: 31825824
DOI: https://doi.org/10.1016/j.celrep.2019.11.009
Primary Citation Related Structures:
6ECH, 6ECK

PubMed Abstract:
Pyruvate kinase is an important enzyme in glycolysis and a key metabolic control point. We recently observed a pyruvate kinase liver isoform (PKL) phosphorylation site at S113 that correlates with insulin resistance in rats on a 3 day high-fat diet (HFD) and suggests additional control points for PKL activity....
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Organizational Affiliation:
Department of Cellular and Molecular Physiology, Yale University, New Haven, CT, USA; Department of Systems Biology Institute, Yale University, New Haven, CT, USA.
View More

Biological Assembly 1



Explore in 3D: Structure | Sequence Annotations | Electron Density | Validation Report | Ligand Interaction (ATP)

Global Symmetry: Dihedral - D2 (Explore in 3D)
Global Stoichiometry: Homo 4-mer - A4

Find Similar Assemblies

Biological assembly 1 assigned by authors and generated by PISA (software)

Biological Assembly Evidence: gel filtration

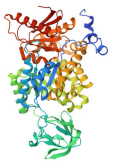
Macromolecule Content

- Total Structure Weight: 245.83 kDa
- Atom Count: 17,686
- Modeled Residue Count: 2,108
- Deposited Residue Count: 2,200
- Unique protein chains: 1

Macromolecules

Find similar proteins by: Sequence | 3D Structure

Entity ID: 1

Molecule	Chains	Sequence Length	Organism	Details	Image
Pyruvate kinase PKLR	A, B, C, D	550	Rattus norvegicus	**Mutation(s):** 0 **Gene Name s:** Pklr **EC:** 2.7.1.40	

UniProt

Find proteins for P12928 *(Rattus norvegicus)* Explore P12928 Go to UniProtKB: P12928

Entity Groups

Sequence Clusters	30% Identity, 50% Identity, 70% Identity, 90% Identity, 95% Identity, 100% Identity
UniProt Group	P12928

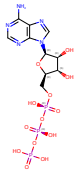
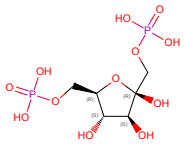
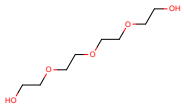
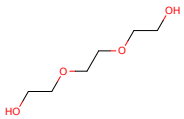
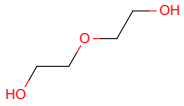
Sequence Annotations Expand

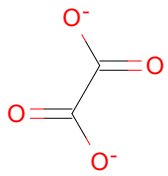
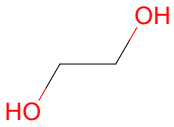
Reference Sequence 6ECH_1



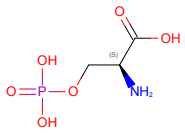
Small Molecules

Ligands 9 Unique

ID	Chains	Name / Formula / InChI Key	2D Diagram	3D Interactions
ATP Download: Instance Coordinates	CB [auth D], J [auth A], Z [auth B]	**ADENOSINE-5'-TRIPHOSPHATE** C_{10} H_{16} N_5 O_{13} P_3 ZKHQWZAMYRWXGA-KQYNXXCUSA-N		Interactions Interactions & Density
FBP Download: Instance Coordinates	E [auth A], LA [auth C], U [auth B], YA [auth D]	**1,6-di-O-phosphono-beta-D-fructofuranose** C_6 H_{14} O_{12} P_2 RNBGYGVWRKECFJ-ARQDHWQXSA-N		Interactions Interactions & Density
PG4 Download: Instance Coordinates	T [auth A]	**TETRAETHYLENE GLYCOL** C_8 H_{18} O_5 UWHCKJMYHZGTIT-UHFFFAOYSA-N		Interactions Interactions & Density
PGE Download: Instance Coordinates	BA [auth B]	**TRIETHYLENE GLYCOL** C_6 H_{14} O_4 ZIBGPFATKBEMQZ-UHFFFAOYSA-N		Interactions Interactions & Density
PEG Download: Instance Coordinates	MB [auth D], S [auth A]	**DI(HYDROXYETHYL)ETHER** C_4 H_{10} O_3 MTHSVFCYNBDYFN-UHFFFAOYSA-N		Interactions Interactions & Density

OXL Download: Instance Coordinates	AA [auth B], DB [auth D], K [auth A], OA [auth C]	**OXALATE ION** C_2 O_4 MUBZPKHOEPUJKR-UHFFFAOYSA-L		Interactions Interactions & Density
EDO Download: Instance Coordinates		**1,2-ETHANEDIOL** C_2 H_6 O_2 LYCAIKOWRPUZTN-UHFFFAOYSA-N		Interactions Interactions & Density
CL Download: Instance Coordinates	NB [auth D]	**CHLORIDE ION** Cl VEXZGXHMUGYJMC-UHFFFAOYSA-M		Interactions Interactions & Density
MG Download: Instance Coordinates		**MAGNESIUM ION** Mg JLVVSXFLKOJNIY-UHFFFAOYSA-N		Interactions Interactions & Density

Non-standard Monomers 1 Unique

ID	Chains	Type	Formula	2D Diagram
SEP	A, B, C, D	L-PEPTIDE LINKING	C_3 H_8 N O_6 P	

Experimental Data & Validation

Experimental Data

Method: X-RAY DIFFRACTION
Resolution: 2.19 Å
R-Value Free: 0.238 (Depositor), 0.241 (DCC)
R-Value Work: 0.181 (Depositor), 0.189 (DCC)
R-Value Observed: 0.184 (Depositor)

Space Group: P 2_1 2_1 2_1

Unit Cell:

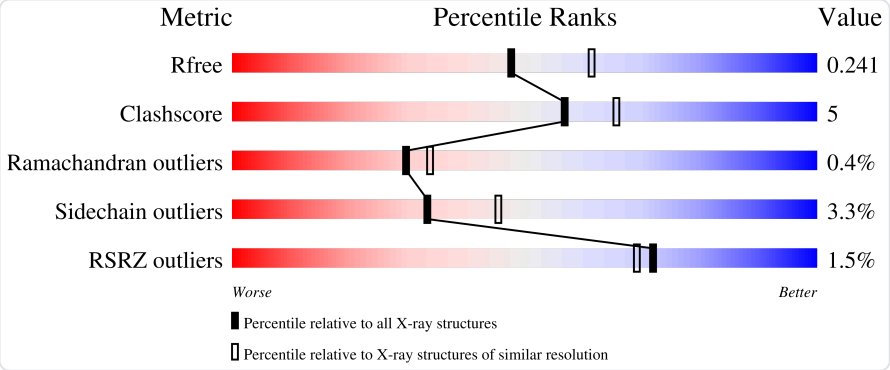
Length (Å)	Angle (°)
a = 86.55	α = 90
b = 108.9	β = 90
c = 296.84	γ = 90

Software Package:

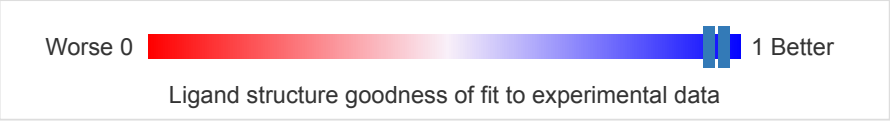
Software Name	Purpose
REFMAC	refinement
HKL-2000	data reduction
Aimless	data scaling
PHASER	phasing
PDB_EXTRACT	data extraction

Structure Validation

View Full Validation Report



Ligand Structure Quality Assessment



Entry History

Deposition Data

Released Date: 2019-12-04
Deposition Author(s): Padyana, A., Tong, S.

Revision History (Full details and data files)

- **Version 1.0: 2019-12-04**
  Type: Initial release
- **Version 1.1: 2019-12-25**
  Changes: Database references
- **Version 1.2: 2020-07-29**
  Type: Remediation
  Reason: Carbohydrate remediation
  Changes: Data collection, Derived calculations, Structure summary
- **Version 1.3: 2023-10-11**
  Changes: Data collection, Database references, Refinement description, Structure summary
- **Version 1.4: 2024-10-09**
  Changes: Structure summary

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Partnerships & Collaborators:

PDB-101 WORLDWIDE PDB PROTEIN DATA BANK EMDataResource NAKB wwPDB Foundation PDB-IHM

RCSB PDB Core Operations are funded by the U.S. National Science Foundation (DBI-2321666), the US Department of Energy (DE-SC0019749), and the National Cancer Institute, National Institute of Allergy and Infectious Diseases, and National Institute of General Medical Sciences of the National Institutes of Health under grant R01GM157729. RCSB PDB uses resources of the National Energy Research Scientific Computing Center (NERSC), a Department of Energy User Facility.